

Models of computation?

mathematically?
 DNA
 Quantum ✓
 classical (PC/cellphone)

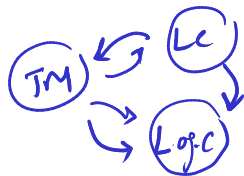
Model

- ① • **Formal, abstract** definition of a computing processor

- ② • Model of computation describes
 - set of allowable **operations** (components) limits
 - cost of respective operations (components) limits
 Analogy's $\rightarrow$ #steps (time), memory (space)

- ③ • There are various models of computation functional
 - Turing machines, Lambda Calculus, Logic circuits
 - Quantum computational model of computation

- ④ • Different models have different **expressive powers**
 - Turing machine is more expressive than finite automata



complexity of
 fun that can be
 simulated

